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Week 6: Atmospheric Seeing and Sky Conditions

Field Notes | Turbulence, the Fried Parameter, and the D/ r_0 *Regime*

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Abstract

Atmospheric seeing is the angular broadening imposed on starlight by refractive index irregularities in the troposphere. Unlike instrumental aberrations, seeing is a property of the air column rather than of the optical system beneath it. This journal quantifies the long-exposure seeing FWHM through the Fried coherence parameter r_0, which emerges from the Kolmogorov turbulence spectrum, and uses the Rayleigh diffraction criterion to define the D/r_0 regime boundary that separates diffraction-limited from seeing-limited operation. At the Melbourne median r_0 of approximately 70 mm, every aperture in the instrument collection examined here falls in the seeing-limited regime. A complementary analysis introduces the sampling ratio η, which maps the angular seeing disk onto the detector pixel grid and connects this journal directly to the Week 8 treatment of sampling theory. The governing equations are instrument-independent: any reader can substitute their own aperture, wavelength, and locally measured r_0 to reproduce all predictions. An Instrument Trade Study quantifies D/r_0, ϑ_R, and η for each standard configuration, and an experimental procedure for direct seeing measurement is fully specified with observing target, site guidance, numbered steps, and quantitative validation criteria.

Keywords: *atmospheric seeing; fried parameter; kolmogorov turbulence; d/r_0 regime; rayleigh criterion; sampling ratio; diffraction limit; wavefront distortion; FWHM; coherence length; site selection; southern victoria*

Nomenclature

Symbol	Definition
θ_{see}	Seeing-limited long-exposure FWHM of the stellar point spread function (radians; converted to arcsec by multiplying by 206265 arcsec rad^-1)
θ_R	Rayleigh criterion angular resolution threshold (radians; same conversion as θ_{see}; equals 138.4 / D^{mm} arcsec at λ = 550 nm)
λ	Imaging wavelength (meters; reference value 550 nm = 5.50e-7 m for broadband visible imaging)
r_0	Fried parameter r_0: the atmospheric coherence length, defined such that the mean-square wavefront phase error across an aperture of diameter r_0 is about 1rad^2.
D	Telescope primary aperture diameter (meters in equations; millimeters in trade study columns where $D/r_0 = D^{mm} / r_0^{mm}$)
η	Sampling ratio: number of pixels per seeing-disk FWHM; $\eta = \theta_{see} / s$, where s is the image scale in arcsec per pixel; Nyquist criterion requires η >= 2 for the profile to be reliably fitted
s	Image scale (arcsec per pixel); $s = 206.265 \times pixel^{um} / FL^{mm}$, where 206.265 = 206265 arcsec rad^-1 / 1000 converts micrometers to millimeters
0.978	Dimensionless Kolmogorov-Fried coefficient in the seeing FWHM relation: derived from the integral of the atmospheric phase structure function over a circular aperture as formalised by Fried (1966) [1]
1.22	Dimensionless Rayleigh coefficient: the first zero of $J_1(x)/x$, where J_1 is the first-order Bessel function; defines the first minimum of the Airy diffraction pattern for a circular aperture
206265	Arcseconds per radian: 1 rad = (360 / 2*pi) x 3600 arcsec = 206265 arcsec rad^-1; used throughout to convert angular quantities from radians to arcseconds
τ_0	Atmospheric coherence time: the interval over which the wavefront phase pattern remains approximately frozen above the aperture; governs the maximum sub-exposure duration for lucky imaging (seconds)
θ_0	Isoplanatic angle: the angular field radius within which a single atmospheric phase correction remains valid; relevant to adaptive optics and multi-star PSF uniformity (arcseconds)
FWHM	Full Width at Half Maximum
SCT	Schmidt-Cassegrain Telescope
Sub-Frame Selector	A sub-frame selector is a tool or module that evaluates and filters individual image exposures ("sub-frames") based on quantitative quality metrics, allowing you to keep only the best data for stacking.

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1. Introduction

Weeks 2 through 5 established the telescope as a collecting and focusing instrument whose performance is governed by aperture, focal length, and focal ratio. Week 6 introduces the first external constraint that no optical design can remove: the atmosphere. Every ground-based observation must pass through a turbulent air column, and that column imposes its own limit on the angular fidelity of the arriving wavefront.

Seeing arises from refractive index fluctuations that distort the wavefront phase, broadening the stellar point spread function. The Fried parameter r_0 quantifies this distortion in a single physically interpretable number.

The ratio D/r_0 then determines whether a given aperture is diffraction-limited or seeing-limited; and the sampling ratio η maps the resulting angular blur onto the detector pixel grid.

All three relationships are instrument-independent, and any reader can apply them to their own telescope, camera, and observing site.

Section 2 defines the experimental design; Sections 3 and 4 derive the governing equations; Section 5 states quantitative predictions; Section 6 applies the full framework to the instrument collection; and Section 7 identifies forward connections to Weeks 7 and 8.

2. Session Configuration and Experimental Design

The experimental goal for Week 6 is to measure atmospheric seeing as an angular property of the air column rather than as a telescope-specific artefact. The key test: if seeing is genuinely atmospheric, the stellar FWHM expressed in arcseconds should remain approximately constant across configurations imaging the same field within the same atmospheric window, while the FWHM expressed in pixels changes in proportion to the image scale s. The session design therefore prioritises high target altitude (minimum air mass), thermal equilibration of the optical train, careful focus, and a stellar-rich field that supports multi-star FWHM statistics.

Primary observing target: Omega Centauri (NGC 5139). Right ascension 13h 26m 47s, declination -47d 28' 46" (J2000). Omega Centauri is the most densely populated globular cluster accessible from the southern hemisphere, with a stellar-rich field approximately 36 arcminutes in diameter. Its stellar density enables robust multi-star FWHM statistics from a single exposure sequence without changing the target field during a configuration swap. Transit altitude from latitude -37.9 degrees South is 90 - |(-37.9) - (-47.5)| = 80.4 degrees, near the zenith and through minimum air mass. Best months for the southern hemisphere: March through June, when the target transits at or before local midnight.

Site selection is guided by two principles: sufficient darkness for star-field acquisition and, more critically, an unobstructed meridian with minimal local thermal sources. Night selection should be guided by the Bureau of Meteorology 500 hPa wind speed forecast; winds below approximately 30 km/h at that level correlate with elevated r_0. Table 1 summarises the recommended site corridors within reach of metropolitan Melbourne.

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Table 1. *Recommended observing site corridors for the Week 6 seeing experiment, ordered by approximate travel time from metropolitan Melbourne. The northeast corridor is preferred for this session because its elevation materially reduces ground-layer turbulence relative to the lower-lying northwest sites.*

Site corridor	Drive from Melbourne	Bortle class	Access	Thermal risk
Northwest (Macedon Ranges area)	~60 min	4-5	Mostly sealed; some unsealed sections; farm gate may apply	Moderate: low-lying farmland releases stored heat after sunset
Northeast (Toolangi/Marysville area)	~75-90 min	3-4	Sealed plus short unsealed section; check seasonal access	Low: elevation 300-700 m reduces ground-layer turbulence
Central Victoria plateau	~120 min	2-3	Partially unsealed; 4WD advisable; confirm access before travel	Very low: elevated, sparsely vegetated; excellent thermal stability

Table 1 shows that the northeast corridor provides the best trade-off between drive time and thermal stability for a systematic seeing measurement. The darker central Victoria sites offer superior sky quality for comparative deep-sky work after the seeing measurements are complete, but the additional drive time reduces the practical session window on weeknights.

Procedure (general method; Askar 71F + ASI585MC on EQM-35 as the worked example throughout):

1. Allow the telescope, mount, and camera to reach ambient temperature for at least 45 minutes before recording any FWHM data. Thermal plumes from a warm tube or mount head mimic poor seeing and will dominate the measurement if equilibration is incomplete. [Worked example: EQM-35 mount with Askar 71F + ASI585MC imaging train.]
2. Perform polar alignment with the ASIAIR Plus built-in routine, targeting a residual below 2 arcminutes. This level of alignment does not produce detectable star elongation in 30-second exposures at 490 mm focal length.
3. Slew to a focus star within 5 degrees of Omega Centauri and above 45 degrees altitude. Achieve focus using the ASIAIR Plus auto-focus routine and record the final step position. Reject focus solutions where the live-view FWHM exceeds 3.0 arcsec; a higher reading indicates either defocus or conditions that will confound the measurement.
4. Capture 30 sub-exposures of 30 seconds each with the ASI585MC at gain 200. Enable autoguiding via the ZWO 30F5 + ASI220MM Mini guide scope and reject any frame where guiding RMS exceeds 1.5 arcsec during the sub, as elongated stars cannot be used for isotropic FWHM analysis.
5. Load the 30 frames into PixInsight Sub-frame Selector. Set the scale entry to 1.22 arcsec per pixel. Run batch measurement; Sub-frame Selector outputs θ_{see} directly in arcseconds when the scale is set. For any other configuration, substitute the image scale s = 206.265 x pixel_um / FL_mm using the OTA focal length and camera pixel size from the respective datasheets.
6. Record the median FWHM and interquartile range across all accepted frames. The median is the session estimate of θ_{see}. A range below 0.3 arcsec indicates a stable atmosphere suitable for configuration comparison. If a second configuration is deployed simultaneously, both must report the same θ_{see} in arcseconds within 10 percent.

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Simultaneous deployment of the Askar 91F APO + ASI533MC Pro on the HEQ5-R Pro at the same site eliminates atmospheric variability as a confound and provides the strongest cross-check of plate-scale calibration: both instruments must report the same θ_{see} in arcseconds despite their differing image scales of 1.22 and 1.27 arcsec per pixel respectively. Sequential comparison within a 30-minute window is valid only when the interquartile range within each individual sequence is below 0.2 arcsec.

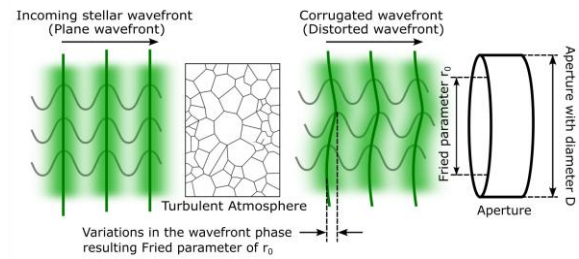

Figure 1. *Stellar wavefront enters the atmosphere as a plane surface and exits with a corrugated phase profile imposed by turbulent refractive index variations. The Fried parameter r_0 marks the lateral scale over which the phase errors remain below 1 rad^2; when the aperture D greatly exceeds r_0, many independent coherence cells contribute to a broadened, time-averaged seeing disk.*

3. The Physics of Atmospheric Turbulence

A productive analogy: consider reading text painted on the bottom of a swimming pool while the surface is disturbed by continuous small ripples.

The text does not move and the observer's eye does not move, yet the changing refractive structure of the water continuously redirects the light path. Atmospheric seeing is the astronomical version of that problem. The source is fixed at stellar distances, the telescope is fixed on a stable mount, yet the arriving wavefront is continuously corrugated by turbulent refractive-index fluctuations in the air column overhead. The distortion is imposed before the photons enter the telescope; no optical refinement inside the instrument can remove it.

The statistical structure of these fluctuations was characterised by Andrei Kolmogorov in 1941 through his analysis of incompressible turbulent flow [2]. The Kolmogorov cascade model describes how kinetic energy injected at large spatial scales, the outer scale of tens to hundreds of meters, breaks down through successively smaller eddies until it dissipates at the inner scale of a few millimeters. Within the inertial subrange, the spatial power spectral density of refractive index fluctuations follows a well-defined power law.

It is this statistical regularity that makes it possible to characterise an entire observing site by a single coherence length [2]. Figure 1 illustrates the sequence from incoming stellar wavefront to atmospheric distortion to the corrugated phase profile entering the telescope aperture [2].

Figure 1 establishes the geometric interpretation on which the quantitative regime analysis of Section 4 rests: the ratio D/r_0 is precisely the number of coherence cells the aperture spans.

When D/r_0 is less than one, the aperture captures a single coherent patch and the PSF is diffraction-limited.

When D/r_0 is much greater than one, the aperture averages many independent phase tilts and the long-exposure PSF converges to a smooth seeing disk whose angular size is set by the atmosphere, not by the telescope optics.

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4. The Fried Parameter and the D/r_0 Regime

David Fried formalised the concept of atmospheric coherence length in his 1966 paper on optical resolution through a randomly inhomogeneous medium.

The Fried parameter r_0 is defined as the aperture diameter at which the integrated Kolmogorov phase variance equals 1 rad^2. Above this threshold, atmospheric phase errors begin to substantially degrade image quality. For apertures smaller than r_0, the wavefront is approximately coherent and the instrument can approach its diffraction limit. The long-exposure seeing disk FWHM for an aperture much larger than r_0 is given by Equation (1) [1].

$$\theta_{\text{see}} = 0.978\frac{\lambda}{r_0} \qquad (1)$$

$\boldsymbol{\theta_{see}}$ = seeing-limited long-exposure FWHM at λ (radians; multiply by 206265 for arcsec)

$\boldsymbol{\lambda}$ = imaging wavelength (meters)

$\mathbf{r_0}$ = Fried coherence parameter (meters)

0.978 = Kolmogorov-Fried proportionality coefficient [1]; derived from the integral of the atmospheric phase structure function over a circular aperture

Equation (1) yields two physically important conclusions. First, θ_{see} is inversely proportional to r_0: halving r_0 doubles the seeing disk. Second, r_0 scales with wavelength. r_0, in this equation, is proportional to wavelength as λ^(6/5), so the net wavelength dependence of θ_{see} is λ^(-1/5): seeing improves modestly at longer wavelengths. Narrowband hydrogen-alpha imaging at 656 nm therefore yields a marginally tighter seeing disk than broadband blue under the same conditions. At Melbourne's median r_0 = 70 mm and λ = 550 nm, Equation (1) gives θ_{see} = 0.978 x 5.50e-7 / 0.070 = 7.70e-6 rad, converting to 7.70e-6 x 206265 = 1.59 arcsec. On poor-to-average nights (r_0 between 30 and 50 mm), θ_{see} rises to 2.2 to 3.5 arcsec; on exceptional nights (r_0 above 120 mm), θ_{see} can fall below 0.9 arcsec [1, 3].

__Table 2.__ Seeing conditions reference: Fried parameter r_0 ranges and corresponding seeing FWHM computed from Equation (1) at λ = 550 nm [1]. Melbourne's median r_0 of approximately 70 mm places typical southern Victorian observing nights in the average-to-good bracket, with ϑ_{see} near 1.6 arcsec on the best median nights.

r_0 (mm)	θ_{see} at 550 nm (arcsec)	Indicative site conditions
20-40	2.8-5.6	Poor to average; turbulent or humid air mass
40-70	1.6-2.8	Average; typical rural or semi-rural dark site
70-120	0.9-1.6	Good; stable high-pressure air, dark rural site
> 120	< 0.9	Excellent; exceptional night or outstanding site

Table 2 anchors the Melbourne reference conditions used throughout this journal. The median r_0 of 70 mm produces θ_{see} near 1.6 arcsec on good nights, rising to 2.5 arcsec or more on average nights. These values set the practical resolution floor for every instrument in the collection, regardless of aperture. Three steps demonstrate the method for any telescope-camera pairing. Step 1: at r_0 = 70 mm and λ = 550 nm, Equation (1) gives θ_{see} = 206265 x 0.978 x 5.50e-7 / 0.070 = 1.59 arcsec.

Step 2: for the Askar 71F + ASI585MC with image scale s = 1.22 arcsec per pixel (206.265 x 2.90 um / 490 mm = 1.22),

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the sampling ratio is $\eta = \theta_{see} / s = 1.59 / 1.22 = 1.30$ pixels per FWHM. Step 3: for the EdgeHD 800 + ASI533MC Pro reduced with s = 0.55 arcsec per pixel, $\eta = 1.59 / 0.55 = 2.89$ pixels per FWHM. The atmosphere is identical in both cases; the pixel representation differs by a factor of 2.2 because the image scale differs by that same ratio. That distinction is the governing observation of the D/r_0 analysis: the same angular PSF, expressed on very different detector footprints.

The diffraction-limited resolution of a circular aperture is given by the Rayleigh criterion, Equation (2). Equation (2), placed alongside Equation (1), quantifies precisely how far any given aperture is from its theoretical diffraction limit under a specific atmospheric condition.

$$\theta_R = 1.22\frac{\lambda}{D} \qquad (2)$$

$\boldsymbol{\theta_R}$ = Rayleigh criterion angular resolution (radians)

$\boldsymbol{\lambda}$ = imaging wavelength (meters)

1.22 = first zero of $J_1(x)/x$; defines the first minimum of the Airy diffraction pattern for a circular aperture [4]

D = telescope primary aperture diameter (meters)

Comparing Equations (1) and (2) directly defines the D/r_0 regime boundary. When D is substantially less than r_0, θ_R exceeds θ_{see}; the wavefront across the aperture is approximately plane and the PSF is dominated by diffraction. When D substantially exceeds r_0, θ_R is far smaller than θ_{see}; the aperture spans many coherence cells and the long-exposure PSF is dominated by the seeing disk. In this seeing-limited regime,

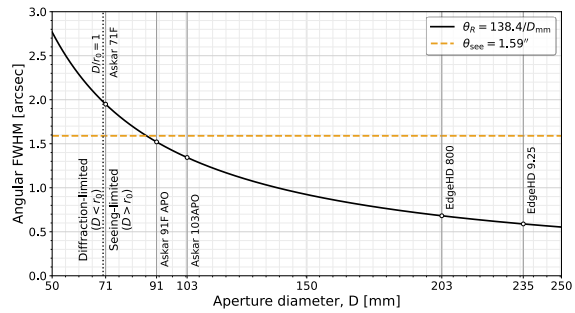

Figure 2. *Seeing-limited FWHM ϑ_{see} (amber dashed, constant at 1.59 arcsec for r_0 = 70 mm, λ = 550 nm) and diffraction-limited Rayleigh FWHM ϑ_R (black solid, decreasing) plotted against aperture D. The dotted vertical line at D = 70 mm marks the regime boundary D/r_0 = 1. All five OTA apertures lie to the right of this boundary at Melbourne's median r_0.*

increasing aperture further does not narrow the stellar profile in a long-exposure image. The aperture advantage then shifts from angular resolution to photon-collecting area: extended objects benefit from additional photons, but stellar point sources do not gain resolution without adaptive correction Equations (1), (2) [1, 4].

The complementary quantity $\eta = \theta_{see} / s$ maps the angular seeing disk onto the detector pixel grid. When η falls below the Nyquist threshold of 2 pixels per FWHM, the stellar profile is under-sampled: the curve-fitting algorithm in Subframe Selector has too few data points to constrain the profile width reliably, and the fitted FWHM is systematically biased upward relative to the true atmospheric value. The D/r_0 ratio and the sampling ratio η together characterise both the atmospheric and the detector constraints on a given configuration, and both are needed to correctly interpret a Subframe Selector result.

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Table 3. *D/r_0 regime analysis and seeing-disk sampling for all standard configurations. ϑ_{see} = 1.59 arcsec for all rows (Equation (1), r_0 = 70 mm, λ = 550 nm). ϑ_R = 138.4 / D^{mm} arcsec (Equation (2), λ = 550 nm). η = 1.59 / s. $D/r_0 = D^{mm}$ / 70. Sampling status categories: Severely under-sampled (η < 1.5); Under-sampled (1.5 <= η < 2.0); Nyquist-met (2.0 <= η < 4.0); Oversampled (η >= 4.0). The same method applies to any telescope-camera combination.*

Configuration (worked example)	D (mm)	D/r_0	θ_R (arcsec)	s (arcsec/px)	η / Sampling status
Askar 71F + ASI585MC	71	1.01	1.95	1.22	1.30 / Severely under-sampled
Askar 91F APO + ASI533MC Pro	91	1.30	1.52	1.27	1.25 / Severely under-sampled
Askar 103APO + ASI533MC Pro (native)	103	1.47	1.34	1.11	1.43 / Under-sampled
Askar 103APO + ASI533MC Pro (0.8x reduced)	103	1.47	1.34	1.38	1.15 / Severely under-sampled
EdgeHD 800 + ASI533MC Pro (0.7x reduced)	203	2.90	0.68	0.55	2.89 / Nyquist-met
EdgeHD 800 + ASI533MC Pro (native)	203	2.90	0.68	0.38	4.18 / Oversampled
EdgeHD 9.25 + ASI462MC (0.7x reduced)	235	3.36	0.59	0.36	4.42 / Oversampled
EdgeHD 9.25 + ASI462MC (native)	235	3.36	0.59	0.25	6.36 / Strongly oversampled

Figure 2 shows θ_{see} and θ_R as functions of aperture D at Melbourne's median r_0 = 70 mm, with all five OTA apertures marked [1, 4].

Figure 2 makes the practical consequence unambiguous: at Melbourne's median r_0, the entire aperture range in the collection sits to the right of the regime boundary, with the Askar 71F closest to the threshold and the larger SCTs operating deepest in the seeing-limited domain. The formal takeaway from Figure 2 is that, under median southern Victorian seeing, increasing aperture below the adaptive-correction threshold usually improves photon collection and sampling leverage before it improves long-exposure stellar resolution. Specific D/r_0 values and their sampling consequences for every standard configuration are quantified in Table 3.

5. Predictions and Validation Criteria

The following predictions are computable from the governing equations and instrument specifications before any observing session is executed. All predicted FWHM values in arcseconds use θ_{see} = 1.59 arcsec, derived from Equation (1) at Melbourne's median r_0 = 70 mm and λ = 550 nm. FWHM values in pixels derive from $\eta = \theta_{see}/s$ where s = 206.265 x pixel_um / FL_mm for each configuration.

For the primary worked example, the Askar 71F + ASI585MC (s = 1.22 arcsec per pixel): predicted θ_{see} = 1.59 arcsec, predicted η = 1.59 / 1.22 = 1.30 pixels per FWHM. For the Askar 91F APO + ASI533MC Pro (s = 1.27 arcsec per pixel), predicted η = 1.59 / 1.27 = 1.25 pixels per FWHM. For the EdgeHD 800 + ASI533MC Pro reduced (s = 0.55 arcsec per pixel), predicted η = 1.59 / 0.55 = 2.89 pixels per FWHM.

The refractor configurations at η < 2 are under-sampled relative to the Nyquist threshold. Sub-frame Selector computes stellar FWHM through Gaussian or Moffat profile fitting to continuous pixel data, returning results with sub-pixel precision; the measurement is not limited by integer-pixel quantisation. However, when a stellar profile spans fewer than 2 pixels per FWHM, the fitting algorithm has too few data points to constrain the profile width reliably. The consequence is systematic upward bias in the fitted FWHM: the reported value will consistently overestimate the true θ_{see}, not because the atmosphere worsened but because the profile fitting is data-limited. For the Askar 71F at η = 1.30, the reported FWHM may therefore read 10 to 20 percent higher than the true seeing value under median conditions. This bias grows as η decreases and is the reason why the EdgeHD configurations, despite sampling the

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same atmosphere, produce more reliable FWHM statistics: their finer image scales keep η comfortably above 2.

An important epistemic point: what Sub-frame Selector reports is the delivered image FWHM, which integrates all blur sources including optical aberration, focus error, guiding residuals, thermal plumes, and the atmosphere. The inferred atmospheric seeing θ_{see} is only obtainable by eliminating or bounding the other contributors. The validation criteria below are therefore not simply quality thresholds; they are the conditions that must be met before the measured FWHM can be treated as an estimate of θ_{see} rather than of the full delivered image quality.

Confirmation criterion for a simultaneous two-configuration comparison: median FWHM in arcseconds should agree within 10 percent after excluding elongated stars (aspect ratio above 1.3) and saturated stars. For a sequential comparison within a 30-minute window, agreement within 15 percent is a more realistic threshold.

A larger discrepancy requires investigation of three sources in order of likelihood. First, focus: defocus broadens the profile systematically across the frame; focus error exceeding 50 microns from the optimal position can inflate the measured FWHM by 0.3 arcsec or more. Second, mount behaviour: elongated stars indicate tracking or guiding errors rather than isotropic atmospheric blur; these frames must be excluded before FWHM statistics are computed. Third, thermal state: a tube or mount head still shedding heat generates local convective plumes even when the free atmosphere is stable. A systematic offset exceeding 0.3 arcsec between the ASIAIR Plus live-view FWHM and Sub-frame Selector output typically indicates a plate scale entry error rather than a genuine atmospheric discrepancy.

Section 6 applies these predictions and validation criteria to the full instrument collection through a structured Instrument Trade Study.

6. Instrument Trade Study

This trade study applies the D/r_0 regime analysis and the sampling ratio $\eta = \theta_{see} / s$ to the full standard configuration set. Aperture diameters and native focal lengths are OTA datasheet values; effective focal lengths for reduced configurations apply the stated reducer factor to the native focal length; pixel sizes are camera datasheet values; image scale s is computed from $s = 206.265 \times pixel^{um} / FL^{mm}$ as introduced in Section 4. The analytical component uses θ_{see} = 1.59 arcsec from Equation (1) at r_0 = 70 mm and λ = 550 nm, and $\theta_R = 138.4 / D^{mm}$ arcsec from Equation (2) at λ = 550 nm.

The same method applies to any telescope-camera combination: substitute the local r0, the aperture in millimeters, and the image scale $s = 206.265 \times pixel^{um} / FL^{mm}$.

6.1 Analytical Component

Table 3 presents D/r_0, θ_R, image scale s, and the combined sampling status for all eight standard configurations. The θ_{see} column is constant at 1.59 arcsec for every row because the atmosphere, not the instrument, sets the seeing disk. The η and sampling status column shows how many pixels that constant disk occupies on each detector and whether the Nyquist threshold of η >= 2 is met.

Table 3 carries three non-obvious results. First, focal reducers do not change D or D/r_0: the Askar 103APO native and reduced rows both

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show D/r_0 = 1.47. A reducer changes the effective focal length but not the collecting aperture; the atmosphere does not know the reducer is present. However, the reducer coarsens the image scale from s = 1.11 to s = 1.38 arcsec per pixel and thereby drops η from 1.43 to 1.15, degrading seeing-disk sampling. Second, all four refractor configurations fall below the Nyquist threshold at median Melbourne conditions. Third, the EdgeHD reduced configurations are the first to cross the Nyquist threshold, not by defeating the atmosphere but by spreading the same seeing disk over more pixels [1].

The practical consequence of η < 2 is more serious than the label 'under-sampled' implies. When Sub-frame Selector fits a Gaussian profile to a stellar image spanning fewer than 2 pixels per FWHM, the fit is systematically biased upward, as established in Section 5. This bias affects the refractors in this collection more than the SCT configurations under median seeing, and it means that Sub-frame Selector FWHM statistics from the Askar 71F, Askar 91F APO, and Askar 103APO should be treated as upper bounds on θ_{see} rather than as unbiased estimates. The direct bridge to Week 8 follows: Week 8 will formalise the Nyquist-Shannon sampling theorem applied to the stellar PSF, and the η values in Table 3 will serve as the starting point for determining whether each configuration can faithfully reconstruct the atmospheric PSF from the recorded pixel data [5]. Under median southern Victorian seeing, increasing aperture below the adaptive-correction threshold usually improves photon collection and sampling leverage before it improves long-exposure stellar resolution. For collecting statistically reliable FWHM measurements with the greatest immunity to fitting bias, the EdgeHD 800 + ASI533MC Pro reduced is the best-suited configuration: its η of 2.89 meets the Nyquist criterion while maintaining the largest aperture among Nyquist-compliant setups.

For wide-field deep-sky imaging under average to below-average conditions, the Askar 91F APO + ASI533MC Pro provides the most efficient match to the atmospheric resolution envelope: its image scale s of 1.27 arcsec per pixel aligns closely with the typical seeing disk, and its D/r_0 of 1.30 means it is only marginally seeing-limited. For compact-object imaging where fine spatial detail is the goal, the EdgeHD 9.25 + ASI462MC reduced (η = 4.42, D/r_0 = 3.36) offers the highest sampling resolution, but demands both steadier-than-median conditions and correspondingly tighter guiding. The least efficient configuration under average Melbourne nights is the EdgeHD 9.25 native (η = 6.36): it oversamples the same seeing disk by a factor of three relative to the Nyquist threshold and requires exceptional conditions before its aperture advantage in resolution becomes accessible.

6.2 Practical Consequences for System Design

Mount and focus discipline become more demanding as image scale s tightens. A residual guiding error of 0.8 arcsec RMS is a minor fraction of the stellar profile for the Askar 71F (s = 1.22 arcsec per pixel) but represents more than two pixels on the EdgeHD 9.25 reduced configuration (s = 0.36 arcsec per pixel). That is the direct link to Week 7: once the atmospheric floor is understood, mount periodic error and autoguiding RMS become the next mechanical constraint that determines whether a fine image scale is useful or merely expensive. The plate scale values in Table 3 determine exactly how

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tightly the mount must perform for each configuration to deliver its intended resolution. Under typical Melbourne conditions (θ_{see} of 2 to 3 arcsec on most nights), wide-field and mid-field refractors are not merely entry instruments. They are efficient matches to the atmospheric resolution available on the majority of nights. Long-focus instruments demand better-than-average seeing and correspondingly tighter guiding before their additional aperture translates into better final images of point sources. The engineering question is therefore not refractor versus SCT in the abstract; it is instrument scale versus atmospheric opportunity on the nights actually available.

7. Forward Notes

Week 7 takes up precisely where Week 6 leaves off. The atmospheric floor established here sets a lower bound on the angular size of the stellar PSF, but it does not determine whether the mount can keep a star on that PSF for the full sub-exposure duration. Mount periodic error and autoguiding RMS impose their own angular blur on the focal plane, and this blur adds in quadrature with the atmospheric term. For configurations with fine image scales such as the EdgeHD reduced setups at 0.55 and 0.36 arcsec per pixel, the tolerable guiding RMS before mount blur exceeds one pixel is 0.55 and 0.36 arcsec respectively, values that define the tracking precision requirements Week 7 will quantify. For the refractor configurations at 1.22 to 1.38 arcsec per pixel, the requirements are proportionally less demanding, explaining why the EQM-35 and HEQ5-R Pro remain viable platforms for those configurations on typical nights.

The D/r_0 analysis treats the atmosphere as a single-layer, stationary process characterised by one number. In practice, r_0 is an altitude-integrated quantity: turbulence layers at different heights and wind speeds each contribute to the total wavefront error. The coherence time τ_0 describes how rapidly the wavefront pattern evolves; it governs the maximum sub-exposure duration for lucky imaging and the closed-loop bandwidth requirement for adaptive optics. The isoplanatic angle θ_0 describes the angular field radius within which a single atmospheric correction remains valid. These extensions to the Fried framework are beyond the Week 6 scope but become directly relevant when planetary lucky imaging or narrowband emission-line work is considered in later content.

The connection to Week 8 (Camera Basics and Sampling Theory) is direct and quantitative. The sampling ratio η introduced here as a prose quantity will be formalised in Week 8 through the Nyquist-Shannon sampling theorem applied to the stellar PSF. The D/r_0 analysis from Week 6 supplies the PSF width predictions that Week 8 will use as its starting point; the two journals are designed to be read as a paired sequence. The most productive observational follow-up to this week's experiment is to repeat the FWHM measurement across several nights spanning a range of atmospheric conditions, building a local seeing envelope: a distribution of measured r_0 values that can later be used to evaluate how often each instrument in the collection can realistically approach its diffraction limit, its Nyquist sampling threshold, or both simultaneously.

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Farzad Farajizadeh (Farzad Faradjizadeh) is an electrical engineer with a doctoral background and a multidisciplinary technical foundation spanning power systems, power electronics, electromagnetic compatibility, and related engineering domains. Through Austronomic, he explores astronomy from a systems-oriented perspective, with a particular emphasis on connecting physical principles, instrumentation, and engineering reasoning in a form accessible to motivated learners.

His broader objective with Austronomic is to develop a structured pathway into astronomy that moves beyond equipment-driven decision making and instead begins with conceptual clarity, technical rigor, and long-term system thinking. This work reflects his interest in building an educational framework in which astronomy can be approached not only as an observational pursuit, but also as a disciplined engineering and scientific practice.

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